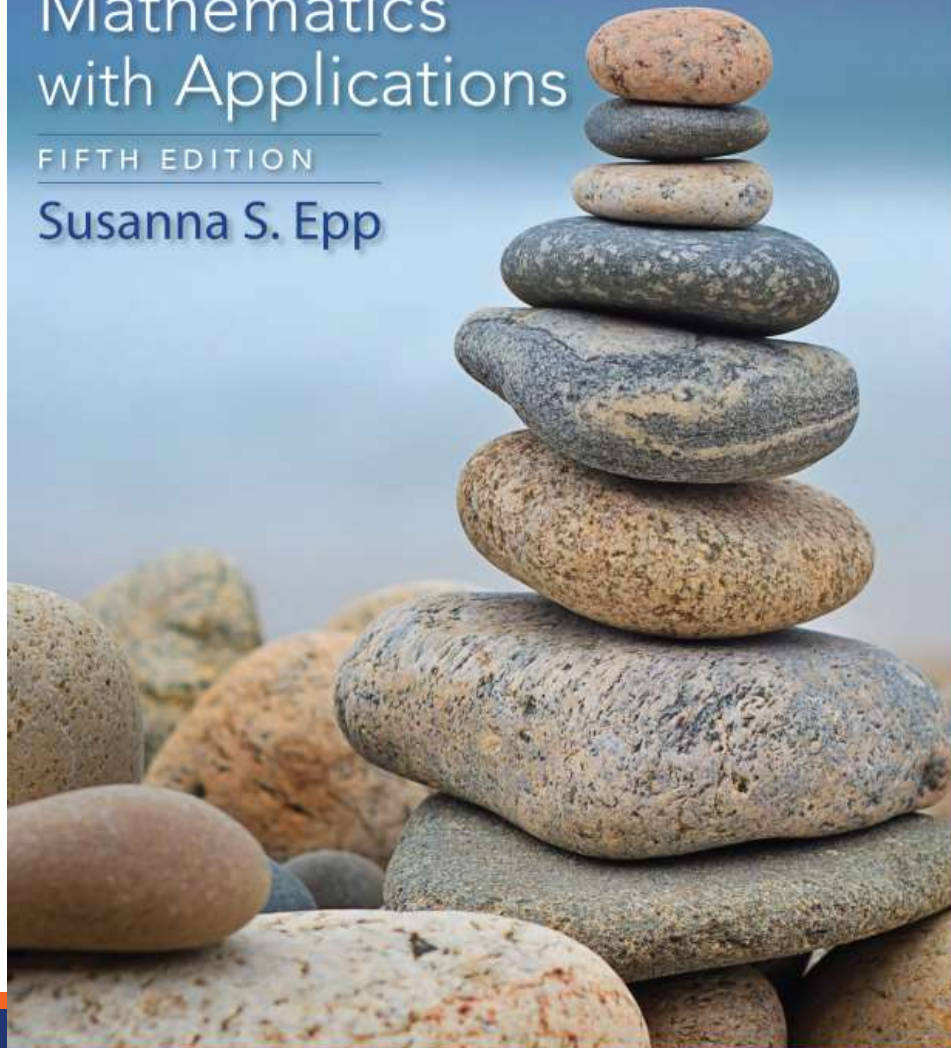


Discrete Mathematics with Applications

FIFTH EDITION

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Faculty of Computer Science

UFCFFS-30-1– Foundations of Computing: Analysis of Algorithms & Algorithm Complexity & Big-O notations

□ Instructor: Duc Minh Vu

Learning Objectives

1. Real-Valued Functions of a Real Variable and Their Graphs
2. Big-O, Big-Omega, and Big-Theta Notations
3. Analysis of Algorithm Efficiency I
4. Exponential and Logarithmic Functions: Graphs and Orders
5. Analysis of Algorithm Efficiency II

Reading

- ❑ Chapter 11 - Discrete Mathematics with Applications 5th Edition
(Susanna's Book)
- ❑ Chapter 3 – Discrete Mathematics and Its Applications 7th Edition
(Kenneth Rosen's book)

Real-Valued Functions of a Real Variable and Their Graphs

Definition

A **real-valued function of a real variable** is a function from one set of real numbers to another. If f is a real-valued function of a real variable, then for each real number x in the domain of f there is a unique corresponding real number $f(x)$. The **graph of f** is the set of all points (x, y) in the Cartesian coordinate plane with the property that x is in the domain of f and $y = f(x)$.

The definition of graph (see Figure 11.1.1) means that for each x in the domain of f :

$$y = f(x) \iff \text{the point } (x, y) \text{ lies on the graph of } f.$$

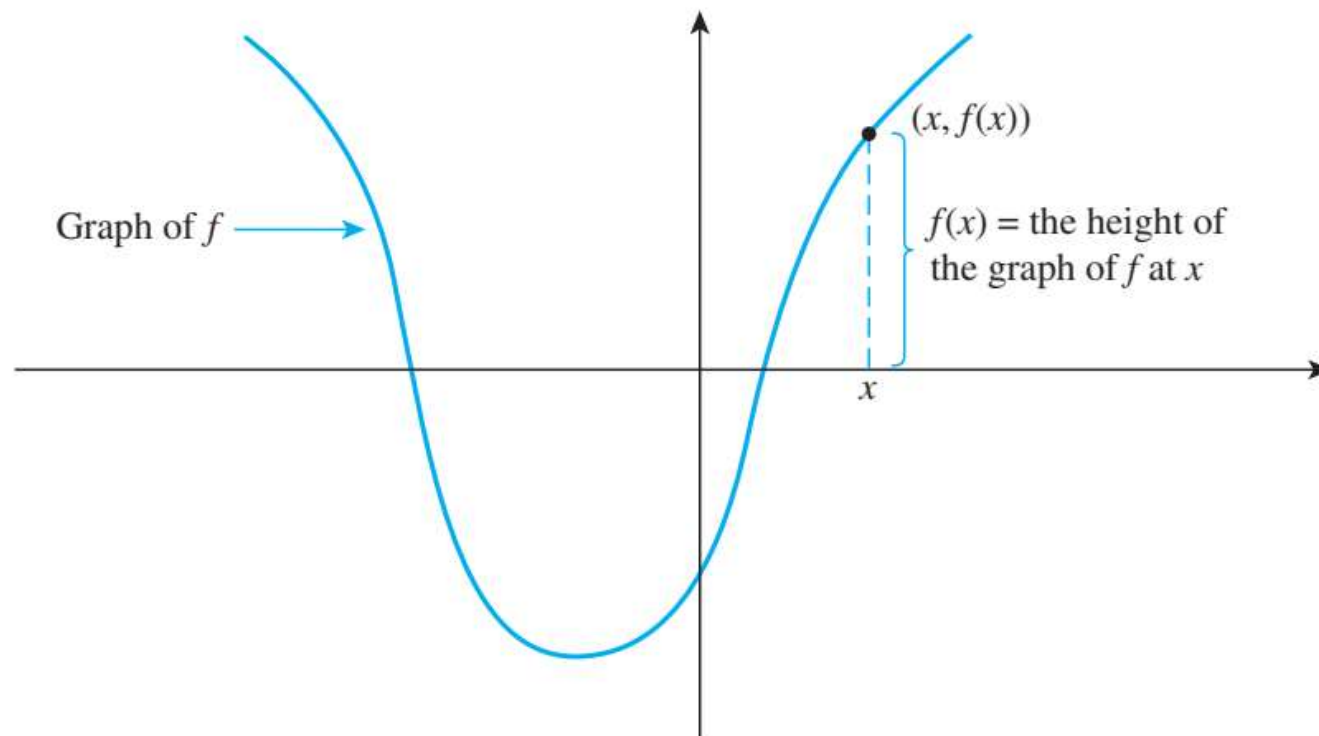


FIGURE 11.1.1 Graph of a Function f

Power Functions

Definition

Let a be any nonnegative real number. Define p_a , the **power function with exponent a** , as follows:

$$p_a(x) = x^a \quad \text{for each nonnegative real number } x.$$

Graphs of Power Functions

Sketch the graphs of the power functions p_0 , $p_{1/2}$, p_1 , and p_2 on the same coordinate axes.

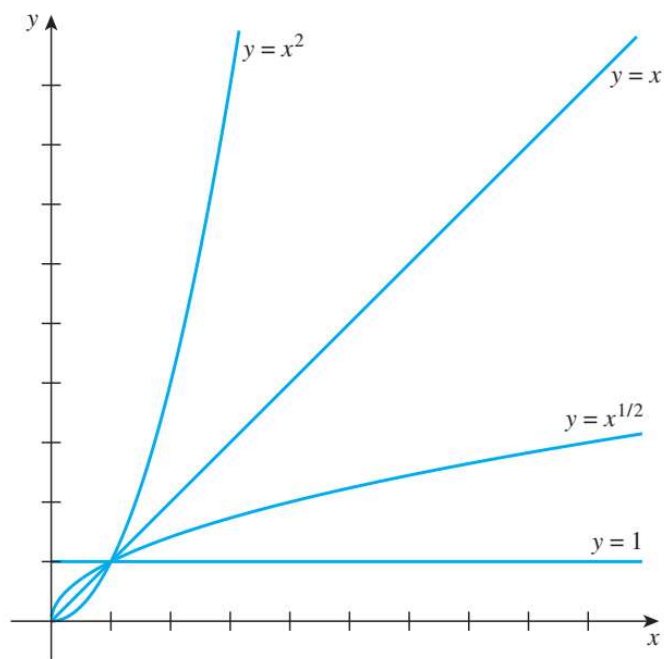


FIGURE 11.1.2 Graphs of Some Power Functions

The Floor Function

Recall that each real number either is an integer itself or sits between two consecutive integers: For each real number x , there exists a unique integer n such that $n \leq x < n + 1$. The floor of a number is the integer immediately to its left on the number line. More formally, the floor function F is defined by the rule

For each real number x ,

$$F(x) = \lfloor x \rfloor$$

= the greatest integer that is less than or equal to x

= the unique integer n such that $n \leq x < n + 1$.

Graph of the Floor Function

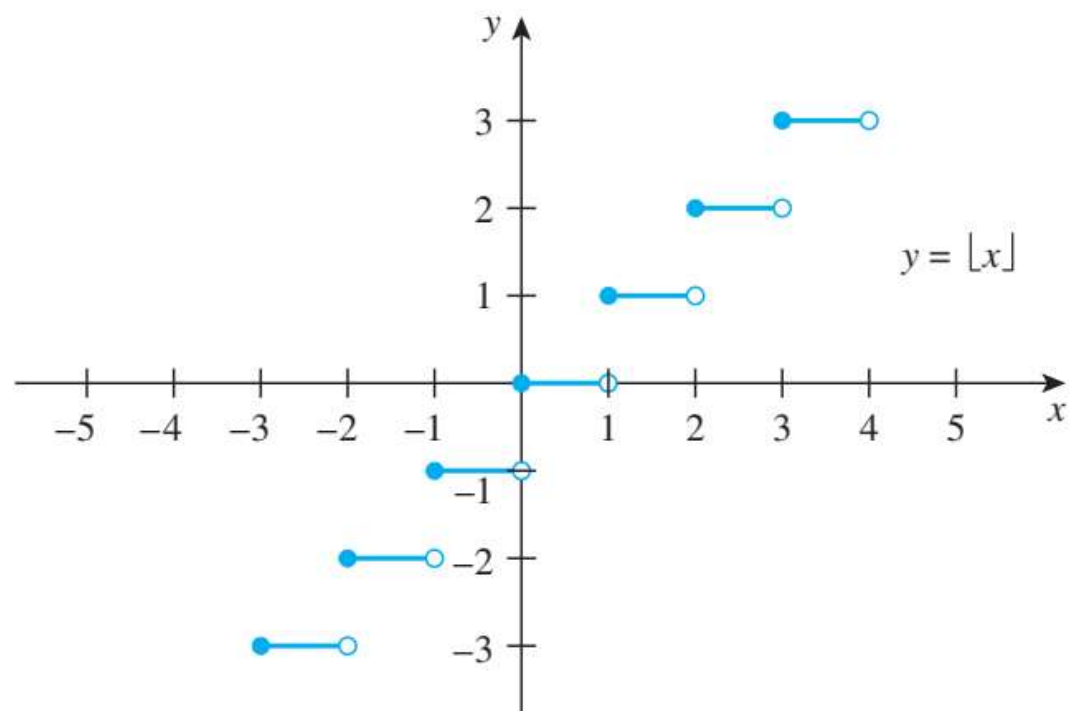


FIGURE 11.1.3 Graph of the Floor Function

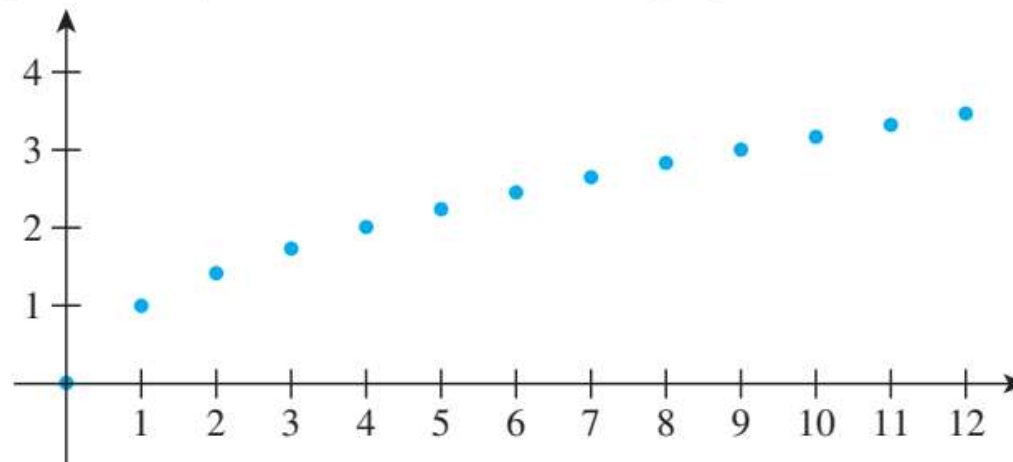
Graphing Functions Defined on Sets of Integers

- Many real-valued functions used in computer science are defined on sets of integers rather than on intervals of real numbers.
- If you know what the graph of a function looks like when it is given by a formula on an interval of real numbers, you can obtain the graph of the function that is defined on the integers in the interval using the same formula by selecting only the points on the known graph whose first coordinates are integers.

Graph of a Function Defined on a Set of Integers

Consider an integer version of the power function $p_{1/2}$. In other words, define a function g by the formula $g(n) = n^{1/2}$ for each nonnegative integer n . Sketch the graph of g .

Solution Look back at the graph of $p_{1/2}$ in Figure 11.1.2. Draw the graph of g by reproducing only those points on the graph of $p_{1/2}$ with integer first coordinates. Thus for each nonnegative integer n , the point $(n, n^{1/2})$ is on the graph of g .



Graph of g where $g(n) = n^{1/2}$ for each nonnegative integer n

Graph of a Multiple of a Function

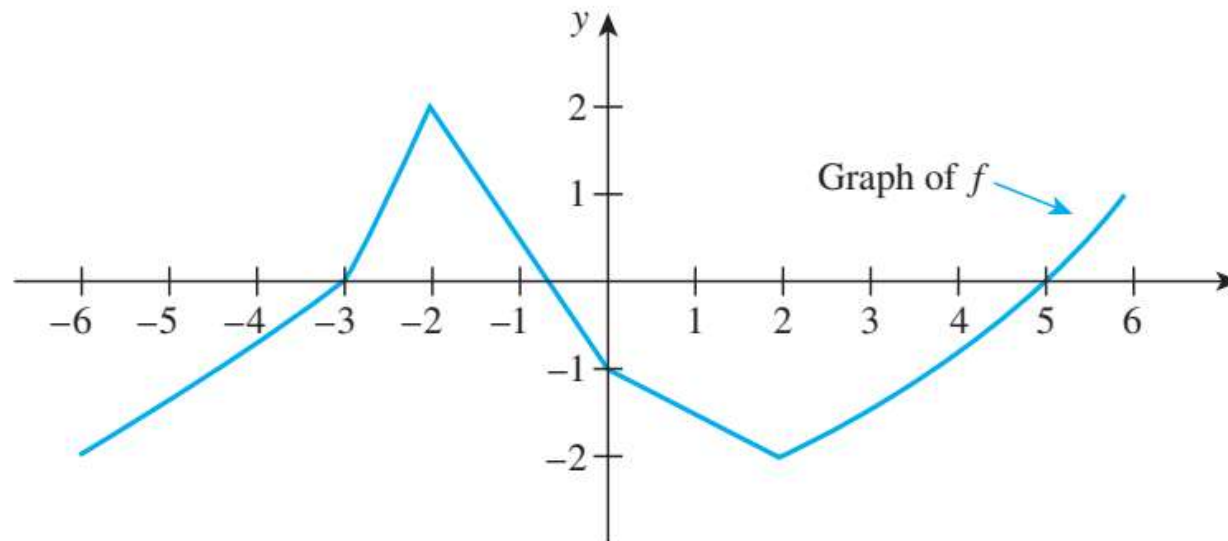
Definition

Let f be a real-valued function of a real variable and let M be any real number. The function Mf , called the **multiple of f by M** or **M times f** , is the real-valued function with the same domain as f that is defined by the rule

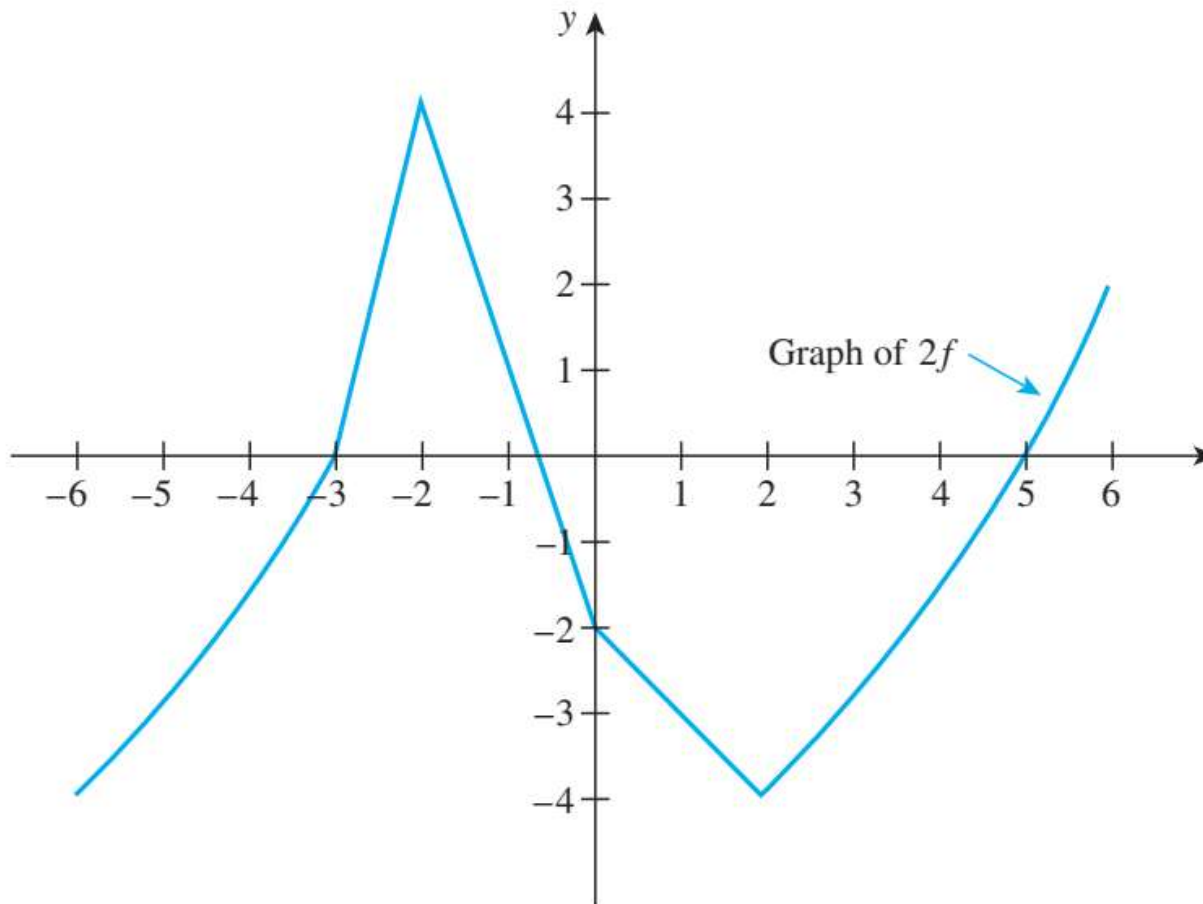
$$(Mf)(x) = M \cdot (f(x)) \quad \text{for each } x \in \text{domain of } f.$$

Graph of a Multiple of a Function

Let f be the function whose graph is shown below. Sketch the graph of $2f$.



Solution At each real number x , you obtain the height of the graph of $2f$ by measuring the height of the graph of f at x and multiplying that number by 2. The result is the following graph. Note that the general shapes of f and $2f$ are very similar, but the graph of $2f$ is “stretched out”: the “highs” are twice as high and the “lows” are twice as low.



Increasing and Decreasing Functions

Definition

Let f be a real-valued function defined on a set of real numbers, and suppose the domain of f contains a set S . We say that f is **increasing on the set S** if, and only if,

for all real numbers x_1 and x_2 in S , if $x_1 < x_2$ then $f(x_1) < f(x_2)$.

We say that f is **decreasing on the set S** if, and only if,

for all real numbers x_1 and x_2 in S , if $x_1 < x_2$ then $f(x_1) > f(x_2)$.

We say that f is an **increasing** (or **decreasing**) **function** if, and only if, f is increasing (or decreasing) on its entire domain.

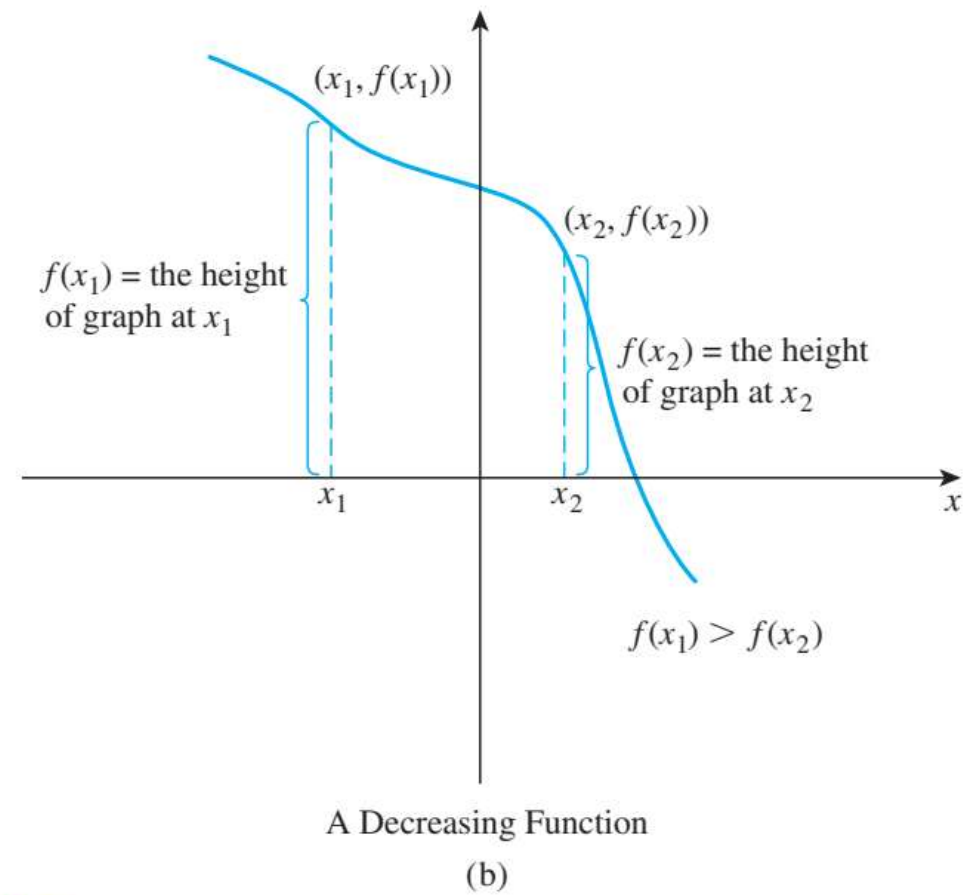
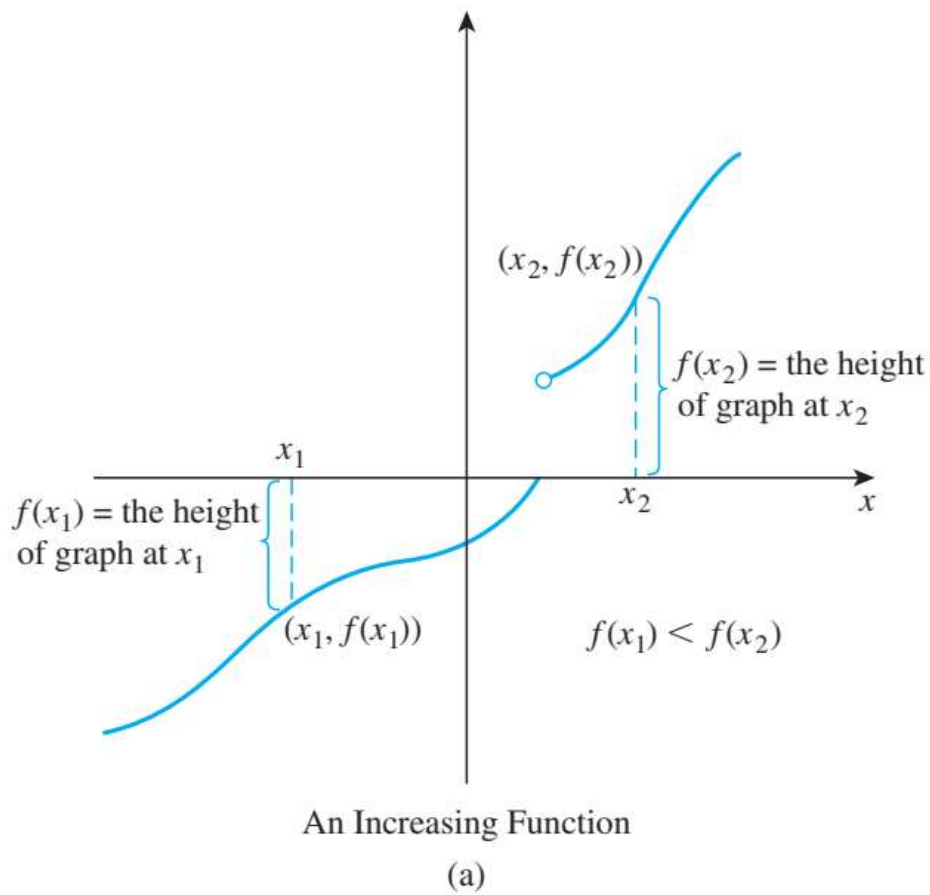


FIGURE 11.1.5

Example 11.1.5

A Positive Multiple of an Increasing Function Is Increasing

Suppose that f is a real-valued function of a real variable that is increasing on a set S of real numbers, and suppose M is any positive real number. Show that Mf is also increasing on S .

Solution Suppose x_1 and x_2 are particular but arbitrarily chosen elements of S such that

$$x_1 < x_2.$$

[We must show that $(Mf)(x_1) < (Mf)(x_2)$.] From the facts that $x_1 < x_2$ and f is increasing, it follows that

$$f(x_1) < f(x_2).$$

Then

$$Mf(x_1) < Mf(x_2),$$

since multiplying both sides of the inequality by a positive number does not change the direction of the inequality. Hence, by definition of Mf ,

$$(Mf)(x_1) < (Mf)(x_2),$$

and, consequently, Mf is increasing on S . 

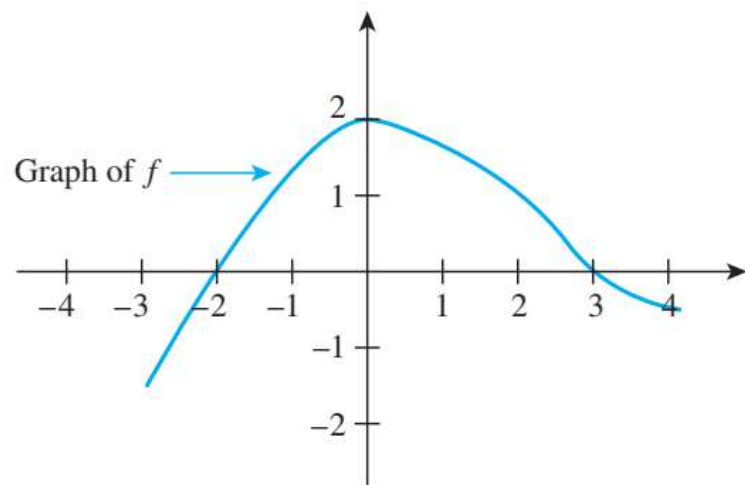
TEST YOURSELF

Answers to Test Yourself questions are located at the end of each section.

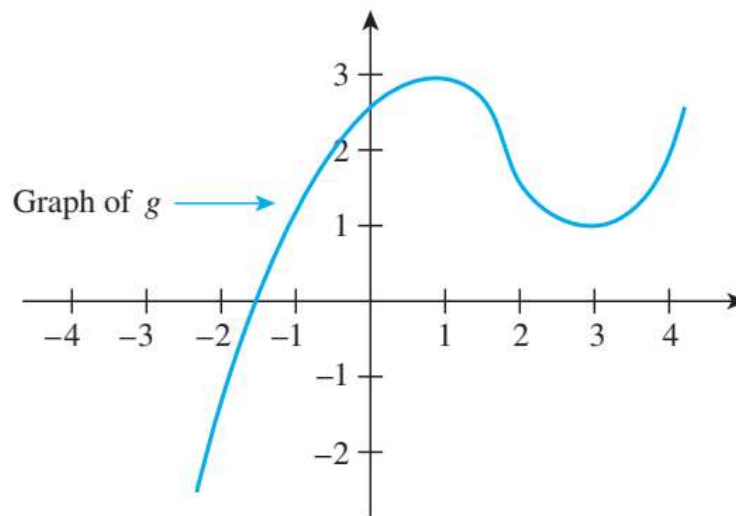
1. If f is a real-valued function of a real variable, then the domain and co-domain of f are both _____.
2. A point (x, y) lies on the graph of a real-valued function of a real variable f if, and only if, _____.
3. If a is any nonnegative real number, then the power function with exponent a , p_a , is defined by _____.
4. Given a function $f: \mathbf{R} \rightarrow \mathbf{R}$ and a real number M , the function Mf is defined by _____.
5. Given a function $f: \mathbf{R} \rightarrow \mathbf{R}$, to prove that f is increasing, you suppose that _____ and then you show that _____.
6. Given a function $f: \mathbf{R} \rightarrow \mathbf{R}$, to prove that f is decreasing, you suppose that _____ and then you show that _____.

EXERCISE SET 11.1*

1. The graph of a function f is shown below.
 - a. Is $f(0)$ positive or negative?
 - b. For what values of x does $f(x) = 0$?
 - c. Find approximate values for x_1 and x_2 so that $f(x_1) = f(x_2) = 1$ but $x_1 \neq x_2$.
 - d. Find an approximate value for x such that $f(x) = 1.5$.
 - e. As x increases from -3 to -1 , do the values of f increase or decrease?
 - f. As x increases from 0 to 4 , do the values of f increase or decrease?



2. The graph of a function g is shown below.
 - a. Is $g(0)$ positive or negative?
 - b. Find an approximate value of x so that $g(x) = 0$.
 - c. Find approximate values for x_1 and x_2 so that $g(x_1) = g(x_2) = 1$ but $x_1 \neq x_2$.
 - d. Find an approximate value for x such that $g(x) = -2$.
 - e. As x increases from -2 to 1 , do the values of g increase or decrease?
 - f. As x increases from 1 to 3 , do the values of g increase or decrease?



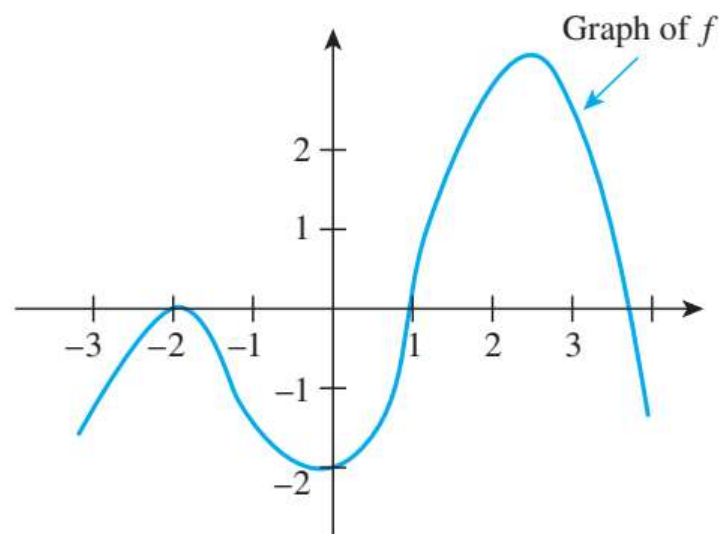
3. Sketch the graphs of the power functions $p_{1/3}$ and $p_{1/4}$ on the same set of axes. When $0 < x < 1$, which is greater: $x^{1/3}$ or $x^{1/4}$? When $x > 1$, which is greater: $x^{1/3}$ or $x^{1/4}$?
4. Sketch the graphs of the power functions p_3 and p_4 on the same set of axes. When $0 < x < 1$, which is greater: x^3 or x^4 ? When $x > 1$, which is greater: x^3 or x^4 ?
5. Sketch the graphs of $y = 2[x]$ and $y = [2x]$ for each real number x . What can you conclude from these graphs?

Sketch a graph for each of the functions defined in 6–9 below.

6. $g(x) = \lceil x \rceil$ for each real number x (Recall that the ceiling of x , $\lceil x \rceil$, is the least integer that is greater than or equal to x . That is, $\lceil x \rceil =$ the unique integer n such that $n - 1 < x \leq n$.)
7. $h(x) = \lceil x \rceil - \lfloor x \rfloor$ for each real number x
8. $F(x) = \lfloor x^{1/2} \rfloor$ for each real number x
9. $G(x) = x - \lfloor x \rfloor$ for each real number x

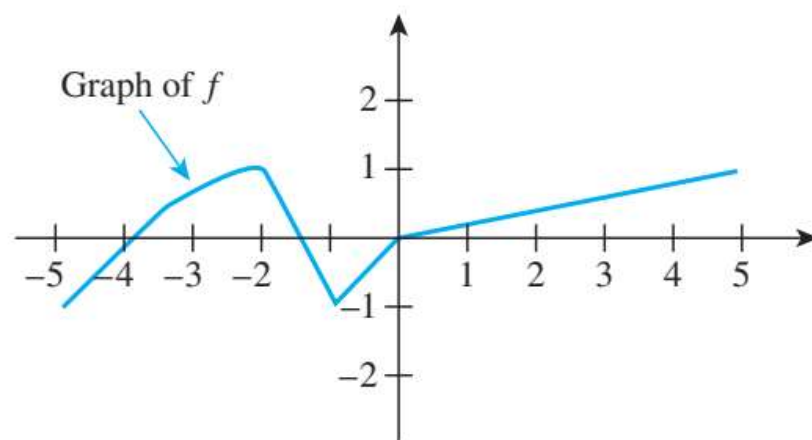
In each of 10–13 a function is defined on a set of integers. Sketch a graph for each function.

10. $f(n) = |n|$ for each integer n
11. $g(n) = (n/2) + 1$ for each integer n
12. $h(n) = \lfloor n/2 \rfloor$ for each integer $n \geq 0$
13. $k(n) = \lfloor n^{1/2} \rfloor$ for each integer $n \geq 0$
14. The graph of a function f is shown below. Find the intervals on which f is increasing and the intervals on which f is decreasing.

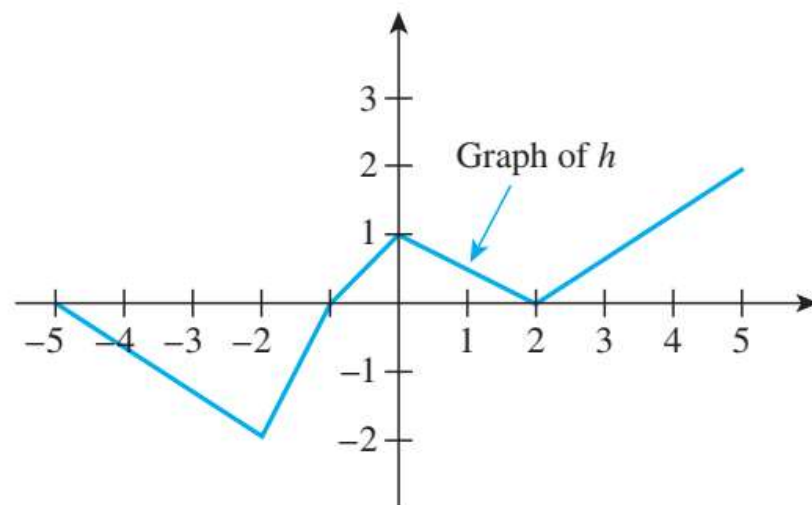


- 15.** Show that the function $f: \mathbf{R} \rightarrow \mathbf{R}$ defined by the formula $f(x) = 2x - 3$ is increasing on the set of real numbers.
- 16.** Show that the function $g: \mathbf{R} \rightarrow \mathbf{R}$ defined by the formula $g(x) = -(x/3) + 1$ is decreasing on the set of real numbers.
- 17.** Let h be the function from $\mathbf{R}$ to $\mathbf{R}$ defined by the formula $h(x) = x^2$ for each real number x
 - a.** Show that h is decreasing on the set of real numbers less than zero.
 - b.** Show that h is increasing on the set of real numbers greater than zero.
- 18.** Let $k: \mathbf{R} \rightarrow \mathbf{R}$ be the function defined by the formula $k(x) = (x - 1)/x$ for each real number $x \neq 0$.
 - a.** Show that k is increasing for every real number $x > 0$.
 - b.** Is k increasing or decreasing for $x < 0$? Prove your answer.

- 19.** Show that if a function $f: \mathbf{R} \rightarrow \mathbf{R}$ is increasing, then f is one-to-one.
- 20.** Given real-valued functions f and g with the same domain D , the sum of f and g , denoted $f + g$, is defined as follows:
- For each real number x , $(f + g)(x) = f(x) + g(x)$.
- Show that if f and g are both increasing on a set S , then $f + g$ is also increasing on S .
- 21. a.** Let m be any positive integer, and define $f(x) = x^m$ for each nonnegative real number x . Use the binomial theorem to show that f is an increasing function.
- b.** Let m and n be any positive integers, and let $g(x) = x^{m/n}$ for each nonnegative real number x . Prove that g is an increasing function.
- Note: The results of exercise 21 are used in the exercises for Sections 11.2 and 11.4.*
- 22.** Let f be the function whose graph follows. Sketch the graph of $3f$.



- 23.** Let h be the function whose graph is shown below. Sketch the graph of $2h$.



- 24.** Let f be a real-valued function of a real variable.
Show that if f is decreasing on a set S and if M is any positive real number, then Mf is decreasing on S .
- 25.** Let f be a real-valued function of a real variable.
Show that if f is increasing on a set S and if M is any negative real number, then Mf is decreasing on S .
- 26.** Let f be a real-valued function of a real variable.
Show that if f is decreasing on a set S and if M is any negative real number, then Mf is increasing on S .

In 27 and 28, functions f and g are defined. In each case sketch the graphs of f and $2g$ on the same set of axes and find a number x_0 so that $f(x) \leq 2g(x)$ for all $x > x_0$. You can find an exact value for x_0 by solving a quadratic equation, or you can find an approximate value for x_0 by using a graphing calculator or computer.

27. $f(x) = x^2 + 10x + 11$ and $g(x) = x^2$ for each real number $x \geq 0$

28. $f(x) = x^2 + 125x + 254$ and $g(x) = x^2$ for each real number $x \geq 0$



Big-O, Big-Omega, and Big-Theta Notations

Big-O, Big-Omega, and Big-Theta Notations

- ❑ The cost and feasibility of implementing a computer algorithm are most affected by the length of computer time and the amount of computer memory the algorithm requires.
- ❑ This chapter concentrates on basic techniques for calculating time efficiency, which is usually the more significant of the two.
- ❑ Best and worst case for an algorithm: e.g. searching an element
- ❑ Focus on the worst case

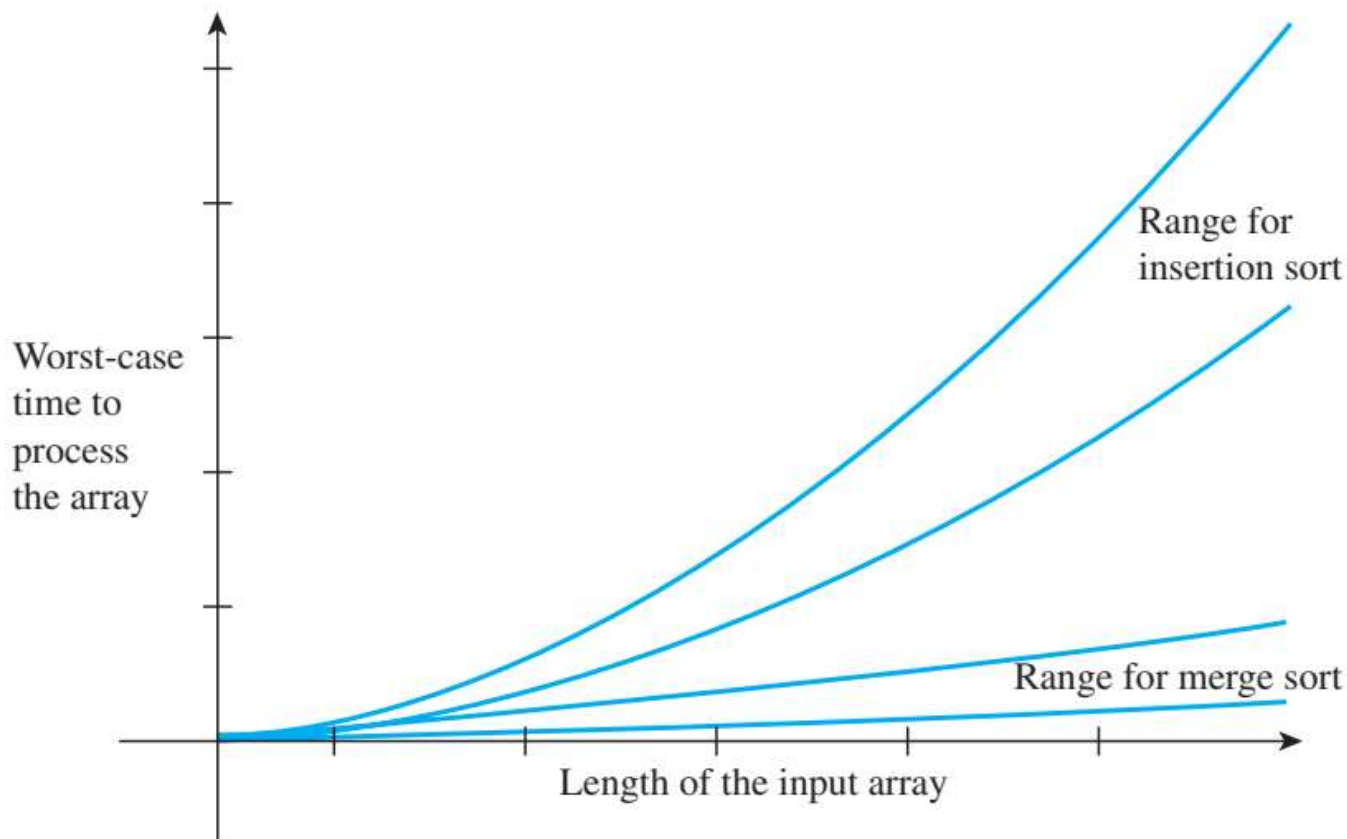


FIGURE 11.2.1

□ For example, sorting a data set with 10 million items could take just a fraction of a second with merge sort but approximately 30 minutes with insertion sort

Big-O, Big-Omega, and Big-Theta Notations

- ❑ O-notation (read “big-O notation”), was introduced by the German mathematician Paul Bachmann in 1894 in a book on analytic number theory, and it was the first to be used to compare efficiencies of algorithms
- ❑ In 1976 Donald Knuth, a pioneer in the analysis of algorithms, introduced the Ω and Θ -notations so that the growth of functions could be compared with greater precision.

The idea of the notations is this: Suppose f and g are real-valued functions of an integer variable n and suppose $g(n) \geq 0$ for every integer n greater than some positive real number.

1. If, for sufficiently large values of n , the values of f are greater than those of a positive multiple of g , then f is of order *at least* g , written “ $f(n)$ is $\Omega(g(n))$.”
2. If, for sufficiently large values of n , the values of f are positive and less than those of a positive multiple of g , then f is of order *at most* g , written “ $f(n)$ is $O(g(n))$.”
3. If, for sufficiently large values of n , the values of f are bounded both above and below by those of positive multiples of g , then f is of order g , written “ $f(n)$ is $\Theta(g(n))$.”

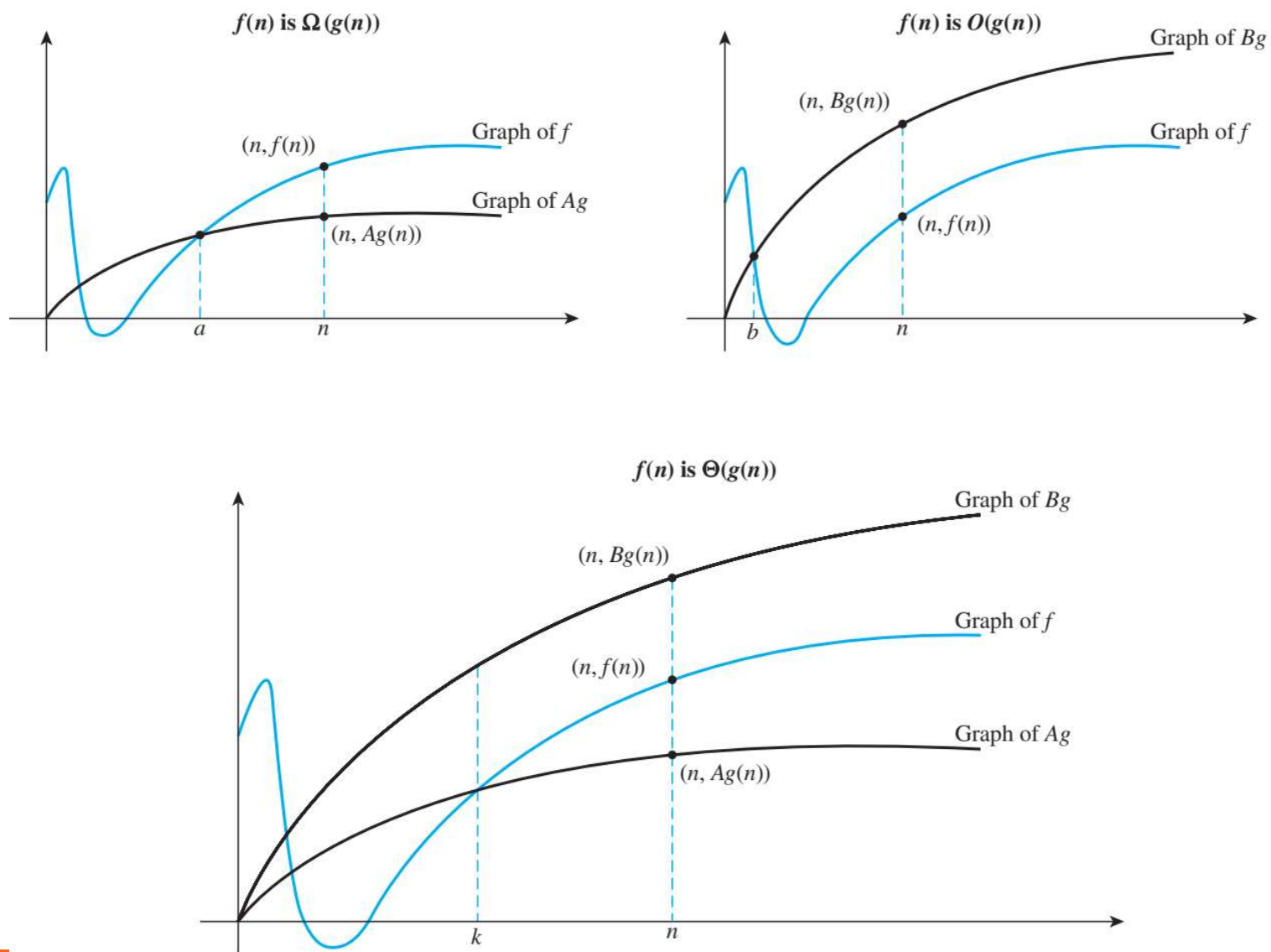


FIGURE 11.2.2

Definition

Let f and g be real-valued functions defined on the same set of nonnegative integers, with $g(n) \geq 0$ for every integer $n \geq r$, where r is a positive real number. Then

1. f is of order at least g , written **$f(n)$ is $\Omega(g(n))$** (f of n is big-Omega of g of n), if, and only if, there exist positive real numbers A and $a \geq r$ such that

$$Ag(n) \leq f(n) \quad \text{for every integer } n \geq a.$$

2. f is of order at most g , written **$f(n)$ is $O(g(n))$** (f of n is big- O of g of n), if, and only if, there exist positive real numbers B and $b \geq r$ such that

$$0 \leq f(n) \leq Bg(n) \quad \text{for every integer } n \geq b.$$

3. f is of order g , written **$f(n)$ is $\Theta(g(n))$** (f of n is big-Theta of g of n), if, and only if, there exist positive real numbers A , B , and $k \geq r$ such that

$$Ag(n) \leq f(n) \leq Bg(n) \quad \text{for every integer } n \geq k.$$

Example 11.2.1 Translating to Θ -Notation

Use Θ -notation to express the statement

$$4n^6 \leq 17n^6 - 45n^3 + 2n + 8 \leq 30n^6 \quad \text{for every integer } n \geq 3.$$

Solution Let $A = 4$, $B = 30$, and $k = 3$. Then the statement translates to

$$An^6 \leq 17n^6 - 45n^3 + 2n + 8 \leq Bn^6 \quad \text{for every integer } n \geq k.$$

So, by definition of Θ -notation,

$$17n^6 - 45n^3 + 2n + 8 \quad \text{is} \quad \Theta(n^6).$$



Example 11.2.2 Translating to Ω - and O -Notations and Deducing Θ -Notation

a. Use Ω -notation to express the statement

$$\frac{11}{4}n^2 \leq 3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \quad \text{for every integer } n \geq 2.$$

b. Use O -notation to express the statement

$$0 \leq 3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \leq 6n^2 \quad \text{for every integer } n \geq 1.$$

c. Justify the statement: $3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2$ is $\Theta(n^2)$.

Solution

a. Let $A = \frac{11}{4}$ and $a = 2$. The statement in (a) translates to

$$An^2 \leq 3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \quad \text{for every integer } n \geq a.$$

So, by definition of Ω -notation,

$$3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \text{ is } \Omega(n^2).$$

b. Let $B = 6$ and $b = 1$. The statement in (b) translates to

$$0 \leq 3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \leq Bn^2 \quad \text{for every integer } n \geq b.$$

So, by definition of O -notation,

$$3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \text{ is } O(n^2).$$

- c. Let $A = \frac{11}{4}$, $a = 2$, $B = 6$, and $b = 1$, and let k be the larger of a and b . Then when $n \geq k$, both inequalities in parts (a) and (b) are satisfied, and so

$$An^2 \leq 3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \leq Bn^2 \quad \text{for every integer } n \geq k.$$

Hence by definition of Θ -notation,

$$3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \quad \text{is} \quad \Theta(n^2).$$



Theorem 11.2.1 Relation among O -, Ω -, and Θ -Notations

If f and g are real-valued functions defined on the same set of nonnegative integers, and if $f(n) \geq 0$ and $g(n) \geq 0$ for every integer $n \geq r$, where r is a positive real number, then $f(n)$ is $\Theta(g(n))$ if, and only if, $f(n)$ is $\Omega(g(n))$ and $f(n)$ is $O(g(n))$.

Orders of Power Functions

Theorem 11.2.2 For any positive rational numbers r and s and any integer $n \geq 1$,

$$\text{if } r \leq s, \text{ then } n^r \leq n^s.$$

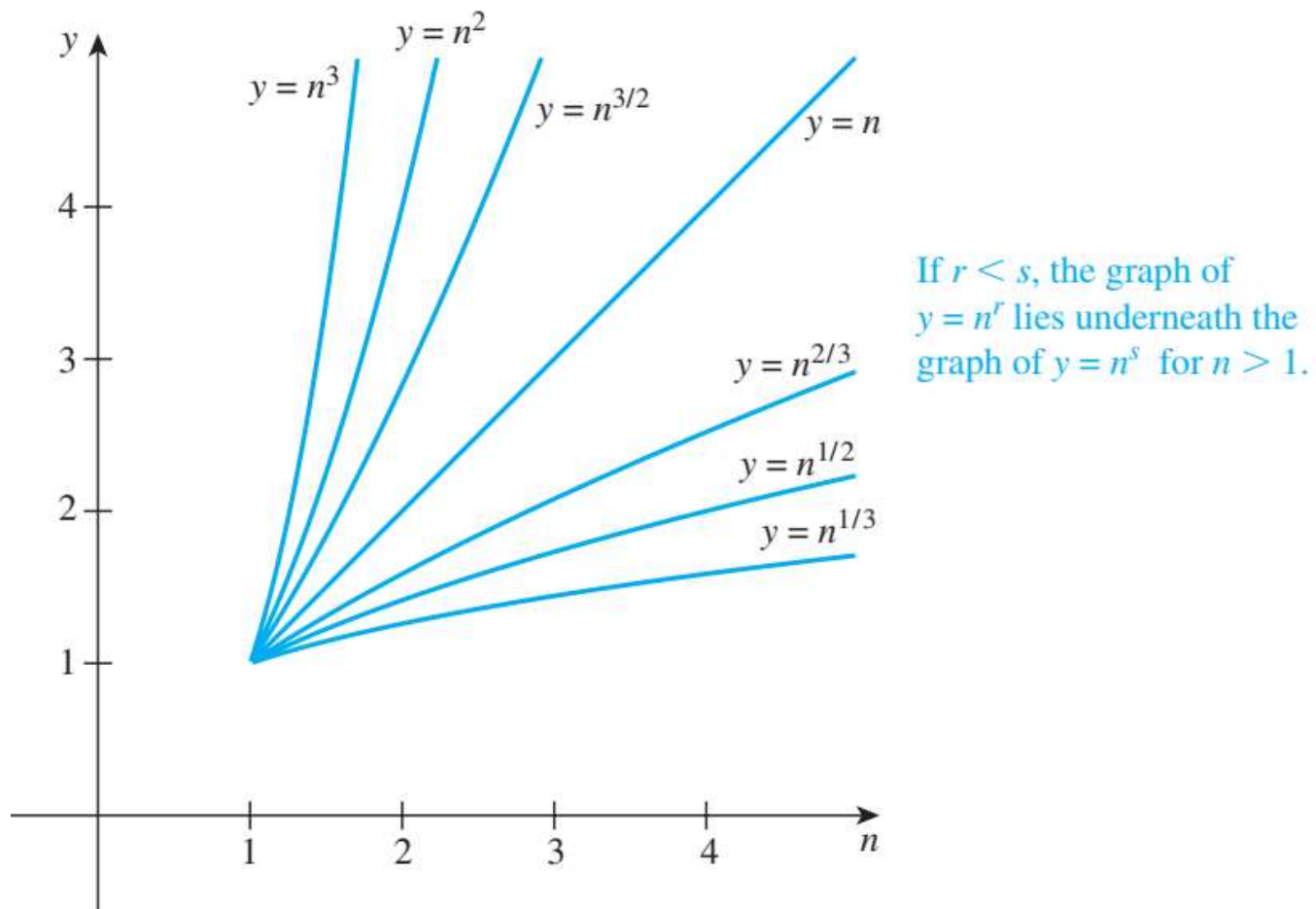


FIGURE 11.2.3 Graphs of Powers of n for $n \geq 1$

Example 11.2.3

Finding a Big-Omega and a Big-O for a Polynomial Function with Nonnegative Coefficients

- a. Show that $15n^3 + 11n^2 + 9$ is $\Omega(n^3)$.
- b. Without using part (a), show that $15n^3 + 11n^2 + 9$ is $O(n^3)$.

Solution

- a. To show that $15n^3 + 11n^2 + 9$ is $\Omega(n^3)$, you need to show that $15n^3 + 11n^2 + 9$ is greater than or equal to a positive multiple of n^3 for all values of n that are sufficiently large. Now

$$15n^3 \leq 15n^3 + 11n^2 + 9 \quad \text{for every integer } n \geq 1$$

because when $n \geq 1$, $11n^2 + 9$ is positive. Thus you can let $A = 15$ and $a = 1$ to obtain

$$An^3 \leq 15n^3 + 11n^2 + 9 \quad \text{for every integer } n \geq a,$$

and conclude, by definition of Ω -notation, that $15n^3 + 11n^2 + 9$ is $\Omega(n^3)$.

- b. To show that $15n^3 + 11n^2 + 9$ is $O(n^3)$, you need to show that $15n^3 + 11n^2 + 9$ is greater than or equal to 0 and less than or equal to some positive multiple of n^3 for all values of n that are sufficiently large. First note that because all terms of $15n^3 + 11n^2 + 9$ are positive,

$$0 \leq 15n^3 + 11n^2 + 9 \quad \text{for every integer } n \geq 1.$$

Next observe that for every integer $n \geq 1$,

$$\begin{aligned} 15n^3 + 11n^2 + 9 &\leq 15n^3 + 11n^3 + 9n^3 && \text{because, by Theorem 11.2.2,} \\ &&& \text{since } n \geq 1, \text{ then } n \leq n^3 \text{ and} \\ &&& 1 \leq n^3, \text{ and so } 11n \leq 11n^3 \\ &&& \text{and } 9 \leq 9n^3 \\ &= 35n^3 && \text{because } 15 + 11 + 9 = 35. \end{aligned}$$

Thus, by transitivity of order and equality,

$$0 \leq 15n^3 + 11n^2 + 9 \leq 35n^3 \quad \text{for every integer } n \geq 1.$$

Let $B = 35$ and $b = 1$. Then

$$0 \leq 15n^3 + 11n^2 + 9 \leq Bn^3 \quad \text{for every integer } n \geq b,$$

and so, by definition of O -notation,

$$15n^3 + 11n^2 + 9 \text{ is } O(n^3).$$

Example 11.2.4

**Finding a Big-Omega and a Big-O for a Polynomial Function
with Some Negative Coefficients**

- a. Show that $n^4 - 5n - 8$ is $\Omega(n^4)$.
- b. Show that $n^4 - 5n - 8$ is $O(n^4)$.

Example 11.2.5

Finding a Big-Theta for a Polynomial Function

- a. Show that $15n^3 + 11n^2 + 9$ is $\Theta(n^3)$.
- b. Show that $n^4 - 5n - 8$ is $\Theta(n^4)$.

Theorem 11.2.3 A Limit on What Can Be Inferred from Big-O

For any function f and positive real numbers r and s with $r < s$,

if $f(n)$ is $O(n^r)$ then $f(n)$ is $O(n^s)$.

Proof: Suppose r and s are real numbers with $r < s$ and f is a function such that $f(n)$ is $O(n^r)$. By definition of O -notation, there exist positive real numbers B and b such that

$$0 \leq f(n) \leq Bn^r \quad \text{for every integer } n \geq b.$$

Now by Theorem 11.2.2,

$$Bn^r \leq Bn^s \quad \text{for every integer } n \geq 1.$$

Let b_1 be the larger of b and 1. Then

$$0 \leq f(n) \leq Bn^s \quad \text{for every integer } n \geq b_1,$$

and thus $f(n)$ is $O(n^s)$.

Example 11.2.7

Showing That One Function Is Not Big-O of Some Other Function

Let $g(n) = n^4 - 5n - 8$. By Example 11.2.4, $g(n)$ is $\Omega(n^4)$. Show that $g(n)$ is not $O(n^r)$ for any positive real number $r < 4$.

Solution

Suppose by way of contradiction that there exists a positive real number r such that $r < 4$ and $g(n)$ is $O(n^r)$. Then there exist positive real numbers B and b such that

$$0 \leq g(n) \leq Bn^r \quad \text{for every integer } n \geq b.$$

Since it is also the case that $g(n)$ is $\Omega(n^4)$, there exist positive real numbers A and a such that

$$An^4 \leq g(n) \quad \text{for every integer } n \geq a.$$

Let t be the larger of a and b . Then

$$An^4 \leq g(n) \leq Bn^r \quad \text{for every integer } n \geq t.$$

Let $\Rightarrow$ stand for the words “which implies that.” It follows from the inequalities above that for every integer $n \geq t$,

$$\begin{aligned}
 & An^4 \leq Bn^r && \text{by transitivity of order} \\
 \Rightarrow & \frac{n^4}{n^r} \leq \frac{B}{A} && \text{by dividing both sides by } An^r \\
 \Rightarrow & n^{4-r} \leq \frac{B}{A} && \text{by algebra} \\
 \Rightarrow & n \leq \sqrt[4-r]{B/A} && \text{by taking the } (4-r)\text{th root of both sides. (*)}
 \end{aligned}$$

Since $r < 4$, then $4 - r > 0$, which implies that $\sqrt[4-r]{B/A}$ is a fixed positive real number. But n can be greater than any fixed number, which implies that condition (*) is contradictory. In other words, the supposition results in a contradiction, and hence it is false. Therefore, for each positive real number r with $r < 4$, $g(n)$ is not $O(n^r)$. ■

Theorem 11.2.4 Showing That a Big-O Relationship Does Not Hold

If f is a real-valued function defined on a set of nonnegative integers and $f(n)$ is $\Omega(n^m)$, where m is a positive integer, then $f(n)$ is not $O(n^p)$ for any positive real number $p < m$.

Theorem 11.2.5 On Polynomial Orders

If m is any integer with $m \geq 0$ and $a_0, a_1, a_2, \dots, a_m$ are real numbers with $a_m > 0$, then $a_m n^m + a_{m-1} n^{m-1} + \dots + a_1 n + a_0$ is $\Theta(n^m)$.

Proof (using limits): Suppose m is an integer with $m \geq 0$ and suppose $a_0, a_1, a_2, \dots, a_m$ are real numbers with $a_m > 0$. Because $\lim_{n \rightarrow \infty} \left(\frac{1}{n^i}\right) = 0$ for every integer $i \geq 1$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{a_m n^m + a_{m-1} n^{m-1} + a_{m-2} n^{m-2} + \dots + a_1 n + a_0}{n^m} \right) \\ = \lim_{n \rightarrow \infty} \left(a_m + \frac{a_{m-1}}{n} + \frac{a_{m-2}}{n^2} + \dots + \frac{a_1}{n^{m-1}} + \frac{a_0}{n^m} \right) \\ = a_m. \end{aligned}$$

By definition of limit, this implies that for any real number $\varepsilon > 0$, there exists an integer K such that

$$a_m - \varepsilon < a_m + \frac{a_{m-1}}{n} + \frac{a_{m-2}}{n^2} + \dots + \frac{a_1}{n^{m-1}} + \frac{a_0}{n^m} < a_m + \varepsilon \quad \text{for every integer } n > K.$$

In particular, when $\varepsilon = \frac{a_m}{2}$, there is an integer k such that

$$a_m - \frac{a_m}{2} < a_m + \frac{a_{m-1}}{n} + \frac{a_{m-2}}{n^2} + \dots + \frac{a_1}{n^{m-1}} + \frac{a_0}{n^m} < a_m + \frac{a_m}{2} \quad \text{for every integer } n > k.$$

Combining like terms and multiplying all parts of the inequality by n^m gives that

$$\left(\frac{a_m}{2}\right)n^m < a_m n^m + a_{m-1} n^{m-1} + \dots + a_1 n + a_0 < \left(\frac{3a_m}{2}\right)n^m \quad \text{for every integer } n > k.$$

Let $A = \frac{a_m}{2}$ and $B = \frac{3a_m}{2}$. Then

$$An^m < a_m n^m + a_{m-1} n^{m-1} + \dots + a_1 n + a_0 < Bn^m \quad \text{for every integer } n > k.$$

Therefore, by definition of Θ -notation,

$$a_m n^m + a_{m-1} n^{m-1} + \dots + a_1 n + a_0 \text{ is } \Theta(n^m).$$

Example 11.2.8

Calculating Polynomial Orders Using the Theorem on Polynomial Orders

Use the theorem on polynomial orders to find orders for the functions given by the following formulas.

a. $f(n) = 7n^5 + 5n^3 - n + 4$ for each positive integer n .

b. $g(n) = \frac{(n-1)(n+1)}{4}$ for each positive integer n .

Solution

a. By direct application of the theorem on polynomial orders, $7n^5 + 5n^3 - n + 4$ is $\Theta(n^5)$.

$$\begin{aligned} \text{b. } g(n) &= \frac{(n-1)(n+1)}{4} \\ &= \frac{1}{4}(n^2 - 1) \\ &= \frac{1}{4}n^2 - \frac{1}{4} \quad \text{by algebra.} \end{aligned}$$

Thus $g(n)$ is $\Theta(n^2)$



Example 11.2.9

An Order for the Sum of the First n Integers

Sums of the form $1 + 2 + 3 + \cdots + n$ arise in the analysis of computer algorithms such as selection sort. Show that for a positive integer variable n ,

$$1 + 2 + 3 + \cdots + n \text{ is } \Theta(n^2).$$

Solution According to the formula for the sum of the first n integers (see Theorem 5.2.1), for each positive integer n ,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Now

$$\frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n \quad \text{by basic algebra.}$$

And, by the theorem on polynomial orders,

$$\frac{1}{2}n^2 + \frac{1}{2}n \quad \text{is} \quad \Theta(n^2).$$

Hence

$$1 + 2 + 3 + \cdots + n \quad \text{is} \quad \Theta(n^2).$$



Theorem 11.2.6 Reciprocal Relationship between Ω - and O -notations

Let f and g be real-valued functions defined on the same set of nonnegative integers, and suppose there is a positive real number r such that $f(n) \geq 0$ and $g(n) \geq 0$ for every integer $n \geq r$. Then:

- a. If $f(n)$ is $\Omega(g(n))$, then $g(n)$ is $O(f(n))$.
- b. If $g(n)$ is $O(f(n))$, then $f(n)$ is $\Omega(g(n))$.

Theorem 11.2.7 Reflexive, Symmetric, and Transitive Properties of Θ -notation

Let f , g , and h be real-valued functions defined on the same set of nonnegative integers, and suppose there is a positive real number r such that $f(n) \geq 0$, $g(n) \geq 0$ and $h(n) \geq 0$, for every integer $n \geq r$. Then:

- a. $f(n)$ is $\Theta(f(n))$.
- b. If $f(n)$ is $\Theta(g(n))$, then $g(n)$ is $\Theta(f(n))$.
- c. If $f(n)$ is $\Theta(g(n))$ and $g(n)$ is $\Theta(h(n))$, then $f(n)$ is $\Theta(h(n))$.

Theorem 11.2.8 Effect of Constants on Order Notations

Let f and g be real-valued functions defined on the same set of nonnegative integers, and suppose there is a positive real number r such that $f(n) \geq 0$ and $g(n) \geq 0$ for every integer $n \geq r$.

Then for every positive real number c :

- a. If $f(n)$ is $\Omega(g(n))$, then $cf(n)$ is $\Omega(g(n))$;
- b. If $f(n)$ is $O(g(n))$, then $cf(n)$ is $O(g(n))$;
- c. If $f(n)$ is $\Theta(g(n))$, then $cf(n)$ is $\Theta(g(n))$.

Theorem 11.2.9 Orders of Sums and Products of Functions

Let f_1, f_2, g_1 , and g_2 be real-valued functions defined on the same set of nonnegative integers, and suppose there is a positive real number r such that $f_1(n) \geq 0$, $f_2(n) \geq 0$, $g_1(n) \geq 0$, and $g_2(n) \geq 0$ for every integer $n \geq r$. Then:

- a. If $f_1(n)$ is $\Theta(g_1(n))$ and $f_2(n)$ is $\Theta(g_2(n))$, then $(f_1(n) + f_2(n))$ is $\Theta(g_1(n) + g_2(n))$.
- b. If $f_1(n)$ is $\Theta(g_1(n))$ and $f_2(n)$ is $\Theta(g_2(n))$, then $(f_1(n)f_2(n))$ is $\Theta(g_1(n)g_2(n))$.
- c. If $f_1(n)$ is $\Theta(g_1(n))$ and $f_2(n)$ is $\Theta(g_2(n))$ and if there is a real number s so that $g_1(n) \leq g_2(n)$ for every integer $n \geq s$, then $(f_1(n) + f_2(n))$ is $\Theta(g_2(n))$.

TEST YOURSELF

1. A sentence of the form “ $Ag(n) \leq f(n)$ for every $n \geq a$ ” translates into Ω -notation as _____.
2. A sentence of the form “ $0 \leq f(n) \leq Bg(n)$ for every $n \geq b$ ” translates into O -notation as _____.
3. A sentence of the form “ $Ag(n) \leq f(n) \leq Bg(n)$ for every $n \geq k$ ” translates into Θ -notation as _____.
4. When $n \geq 1$, n _____ n^2 and n^2 _____ n^5 .
5. According to the theorem on polynomial orders, if $p(n)$ is a polynomial in n , then $p(n)$ is $\Theta(n^m)$, where m is _____.
6. If n is a positive integer, then $1 + 2 + 3 + \cdots + n$ has order _____.

EXERCISE SET 11.2

1. The following is a formal definition for Ω -notation, written using quantifiers and variables: $f(n)$ is $\Omega(g(n))$ if, and only if, $\exists$ positive real numbers a and A such that $\forall n \geq a$,

$$Ag(n) \leq f(n).$$

- a. Write the formal negation for the definition using the symbols $\forall$ and $\exists$.
 - b. Restate the negation less formally without using the symbols $\forall$ and $\exists$ or the words “for any,” “for every,” or “there exists.”
2. The following is a formal definition for O -notation, written using quantifiers and variables: $f(n)$ is $O(g(n))$ if, and only if, $\exists$ positive real numbers b and B such that $\forall n \geq b$,

$$0 \leq f(n) \leq Bg(n).$$

- a. Write the formal negation for the definition using the symbols $\forall$ and $\exists$.
- b. Restate the negation less formally without using the symbols $\forall$ and $\exists$ or the words “for any,” “for every,” or “there exists.”

3. The following is a formal definition for Θ -notation, written using quantifiers and variables: $f(n)$ is $\Theta(g(n))$ if, and only if, $\exists$ positive real numbers k , A , and B such that $\forall n \geq k$,

$$Ag(n) \leq f(n) \leq Bg(n).$$

- a. Write the formal negation for the definition using the symbols $\forall$ and $\exists$.
- b. Restate the negation less formally without using the symbols $\forall$ and $\exists$ or the words “for any,” “for every,” or “there exists.”

In 4–9, express each statement using Ω -, O -, or Θ -notation.

4. $\frac{1}{2}n \leq n - \left\lfloor \frac{n}{2} \right\rfloor + 1$ for every integer $n \geq 1$.
(Use Ω -notation.)

5. $0 \leq n - \left\lfloor \frac{n}{2} \right\rfloor + 1 \leq n$ for every integer $n \geq 3$.
(Use O -notation.)

6. $n^2 \leq 3n(n-2) \leq 4n^2$ for every integer $n \geq 3$.
(Use Θ -notation.)

7. $\frac{1}{2}n^2 \leq \frac{n(3n-2)}{2}$ for every integer $n \geq 3$.
(Use Ω -notation.)
8. $0 \leq \frac{n(3n-2)}{2} \leq n^2$ for every integer $n \geq 1$.
(Use O -notation.)
9. $\frac{n^3}{6} \leq n^2 \left(\left\lceil \frac{n}{3} \right\rceil - 1 \right) \leq n^3$ for every integer $n \geq 2$.
(Use Θ -notation.)

- 10.** **a.** Show that for any integer $n \geq 1$,
 $0 \leq 2n^2 + 15n + 4 \leq 21n^2$.
- b.** Show that for any integer $n \geq 1$,
 $2n^2 \leq 2n^2 + 15n + 4$.
- c.** Sketch a graph to illustrate the results of parts (a) and (b).
- d.** Use the O - and Ω -notations to express the results of parts (a) and (b).
- e.** What can you deduce about the order of $2n^2 + 15n + 4$?

- 11. a.** Show that for any integer $n \geq 1$,
 $0 \leq 23n^4 + 8n^2 + 4n \leq 35n^4$.
- b.** Show that for any integer $n \geq 1$,
 $23n^4 \leq 23n^4 + 8n^2 + 4n$.
- c.** Sketch a graph to illustrate the result of parts (a) and (b).
- d.** Use the O - and Ω -notations to express the results of parts (a) and (b).
- e.** What can you deduce about the order of $23n^4 + 8n^2 + 4n$?

- 12. a.** Show that for any integer $n \geq 1$,
 $0 \leq 7n^3 + 10n^2 + 3 \leq 20n^3$.
- b.** Show that for any integer $n \geq 1$,
 $7n^3 \leq 7n^3 + 10n^2 + 3$.
- c.** Sketch a graph to illustrate the result of parts (a) and (b).
- d.** Use the O - and Ω -notations to express the results of parts (a) and (b).
- e.** What can you deduce about the order of
 $7n^3 + 10n^2 + 3$?

13. Use the definition of Θ -notation to show that $5n^3 + 65n + 30$ is $\Theta(n^3)$.
14. Use the definition of Θ -notation to show that $n^2 + 100n + 88$ is $\Theta(n^2)$.
15. Use the definition of Θ -notation to show that $\left\lfloor n + \frac{1}{2} \right\rfloor$ is $\Theta(n)$.
16. Use the definition of Θ -notation to show that $\left\lceil n + \frac{1}{2} \right\rceil$ is $\Theta(n)$.
17. Use the definition of Θ -notation to show that $\left\lfloor \frac{n}{2} \right\rfloor$ is $\Theta(n)$. (*Hint: Show that if $n \geq 4$, then $\frac{n}{2} - 1 \geq \frac{1}{4}n$.*)

18. Prove Theorem 11.2.7(b): If f and g are real-valued functions defined on the same set of nonnegative integers and if $f(n) \geq 0$ and $g(n) \geq 0$ for every integer $n \geq r$, where r is a positive real number, then if $f(n)$ is $\Theta(g(n))$, then $g(n)$ is $\Theta(f(n))$.
19. Prove Theorem 11.2.1: If f and g are real-valued functions defined on the same set of nonnegative integers and if $f(n) \geq 0$ and $g(n) \geq 0$ for every integer $n \geq r$, where r is a positive real number, then $f(n)$ is $\Theta(g(n))$ if, and only if, $f(n)$ is $\Omega(g(n))$ and $f(n)$ is $O(g(n))$.
20. Without using Theorem 11.2.4 prove that n^5 is not $O(n^2)$.
21. Prove Theorem 11.2.4: If f is a real-valued function defined on a set of nonnegative integers and $f(n)$ is $\Omega(n^m)$, where m is a positive integer, then $f(n)$ is not $O(n^p)$ for any positive real number $p < m$.



23b. We have $f(n) = 1/5n^2$
since $42n + 8 > 0 \Rightarrow$
take $a = 1$, $A = 1/5$
we have $f(n) = Ag(n)$
with $n \geq a \Rightarrow f(n) = O(g(n))$
by definition

- 22. a.** Use one of the methods of Example 11.2.4 to show that $2n^4 - 90n^3 + 3$ is $\Omega(n^4)$.
b. Show that $2n^4 - 90n^3 + 3$ is $O(n^4)$.
c. Justify the conclusion that $2n^4 - 90n^3 + 3$ is $\Theta(n^4)$.

- 23. a.** Use one of the methods of Example 11.2.4 to show that $\frac{1}{5}n^2 - 42n - 8$ is $\Omega(n^2)$.
b. Show that $\frac{1}{5}n^2 - 42n - 8$ is $O(n^2)$.
c. Justify the conclusion that $\frac{1}{5}n^2 - 42n - 8$ is $\Theta(n^2)$.

- 24. a.** Use one of the methods of Example 11.2.4 to show that $\frac{1}{4}n^5 - 50n^3 + 3n + 12$ is $\Omega(n^5)$.
b. Show that $\frac{1}{4}n^5 - 50n^3 + 3n + 12$ is $O(n^5)$.
c. Justify the conclusion that $\frac{1}{4}n^5 - 50n^3 + 3n + 12$ is $\Theta(n^5)$.

23a) Since $42n + 8 = 50n$
so $f(n) \geq 1/5n^2 - 50n$
For all $n \geq 300$, we have,
 $50n = n^2/6$; so $f(n) \geq$
 $1/5n^2 - 1/6n^2$ or $f(n) \geq$
 $1/30n^2$
Take $a = 300$, $A = 1/30$,
we have $f(n) = \Omega(n^2)$ by definition

Use the theorem on polynomial orders to prove each of the statements in 26–31.

26. $\frac{(n+1)(n-2)}{4}$ is $\Theta(n^2)$

27. $\frac{n}{3}(4n^2 - 1)$ is $\Theta(n^3)$

28. $\frac{n(n-1)}{2} + 3n$ is $\Theta(n^2)$

29. $\frac{n(n-1)(2n+1)}{6}$ is $\Theta(n^3)$

30. $\left[\frac{n(n+1)}{2} \right]^2$ is $\Theta(n^4)$

31. $2(n-1) + \frac{n(n+1)}{2} + 4\left(\frac{n(n-1)}{2}\right)$ is $\Theta(n^2)$

Prove each of the statements in 32–39. Use the theorem on polynomial orders and results from the theorems and exercises in Section 5.2 as appropriate.

32. $1^2 + 2^2 + 3^2 + \cdots + n^2$ is $\Theta(n^3)$

33. $1^3 + 2^3 + 3^3 + \cdots + n^3$ is $\Theta(n^4)$

34. $2 + 4 + 6 + \cdots + 2n$ is $\Theta(n^2)$

35. $5 + 10 + 15 + 20 + 25 + \cdots + 5n$ is $\Theta(n^2)$

36. $\sum_{i=1}^n (4i - 9)$ is $\Theta(n^2)$

37. $\sum_{k=1}^n (k + 3)$ is $\Theta(n^2)$

38. $\sum_{i=1}^n i(i + 1)$ is $\Theta(n^3)$

39. $\sum_{k=3}^n (k^2 - 2k)$ is $\Theta(n^3)$

- 40. a.** Prove: If c is a positive real number and if f is a real-valued function defined on a set of nonnegative integers with $f(n) \geq 0$ for every integer n greater than or equal to some positive real number, then $cf(n)$ is $\Theta(f(n))$.
- b.** Use part (a) to show that $3n$ is $\Theta(n)$.
- 41.** Prove: If c is a positive real number and $f(n) = c$ for every integer $n \geq 1$, then $f(n)$ is $\Theta(1)$.
- 42.** What can you say about a function f with the property that $f(n)$ is $\Theta(1)$?

Use Theorems 11.2.5–11.2.9 and the results of exercises 15–17, 40, and 41 to justify the statements in 43–45.

43. $\left\lfloor \frac{n+1}{2} \right\rfloor + 3n$ is $\Theta(n)$

44. $\frac{n(n-1)}{2} + \left\lfloor \frac{n}{2} \right\rfloor + 1$ is $\Theta(n^2)$

45. $\left\lfloor \frac{n}{2} \right\rfloor + 4n + 3$ is $\Theta(n)$

- 46. a.** Use mathematical induction to prove that if n is any integer with $n \geq 1$, then for every integer $m \geq 1$, $n^m \geq 1$.
- b.** Prove that if n is any integer with $n \geq 1$, then $n^r \leq n^s$ for all integers r and s with $r \leq s$.

- 47. a.** Let x be any positive real number. Use mathematical induction to prove that for every integer $m \geq 1$, if $x \leq 1$ then $x^m \leq 1$.
- H b.** Explain how it follows from part (a) that if x is any positive real number, then for every integer $m \geq 1$, if $x^m > 1$ then $x > 1$.
- c.** Explain how it follows from part (b) that if x is any positive real number, then for every integer $m \geq 1$, if $x > 1$ then $x^{1/m} > 1$.
- d.** Let p, q, r , and s be positive integers, and suppose $p/q > r/s$. Use part (c) and the result of exercise 40 to prove Theorem 11.2.2. In other words, show that for any integer n , if $n > 1$ then $n^{p/q} > n^{r/s}$.

